Optimal versus naive diversification: a replication, and what a tracking-error mandate changes

Carlo Hofer. October 2026. Code and notebooks: github.com/ochofer/optimal-vs-naive-diversification.

I replicate DeMiguel, Garlappi and Uppal (2009) on four of the five datasets the paper takes from Ken French's library (the fifth, FF-3-factor, has no tables in the paper), against bands fixed before the code ran, and then ask their question of a long-only index-tracking mandate on French's 49 industry portfolios from 1979 to 2026: whether the choice of covariance estimator still matters once the portfolio must stay close to its benchmark under a weight bound, a turnover cap, an industry tilt, two exclusions and a beta of one. The replication reproduces 29 of the 30 out-of-sample Sharpe ratios that carry a band of 0.03 and all 30 turnover figures within 25 per cent. Under the mandate the four estimators built from three years of daily returns realise 84.7 to 93.4 basis points a year of tracking error and the sample covariance of 120 monthly returns 95.8, so the estimator still matters by about a tenth of the tracking error, with the daily sample covariance and shrinkage the best of the five and every estimator under-forecasting its own tracking error by 22 to 61 per cent. The tilt alone is associated with an active return of +2 to +7 basis points a year against a standard error of 12 to 13, and a factor attribution indicates that this zero is the sum of factor exposures that cost 9 to 13 basis points and industry returns that paid 12 to 20.

Terms used in this note, each with the one meaning it carries throughout.

1/N: the rule that holds every asset at the same weight and rebalances to it each month. **rule**: any of the portfolio rules compared. **excess return**: a return minus the one-month Treasury bill rate. **Sharpe ratio**: the mean monthly excess return divided by its standard deviation. **turnover**: the sum of absolute weight changes at a rebalance; one-way turnover is half of it, the fraction of the portfolio sold. **estimation window**: the months of returns a rule sees when it forms its weights, 120 in the paper. **estimation error**: the difference between estimated moments and true ones. **out-of-sample**: said of a return earned in the month after the estimation window. **in-sample**: said of a figure computed on the same months the weights were estimated on. **band**: an interval, fixed before the code ran, within which a replicated number counts as matching the published one. **gated**: said of a cell a band applies to. **verdict**: pass or miss for a gated cell; a cell with no band is reported beside the published number and has no verdict. **fragility test**: rerunning a rule after adding small Gaussian noise to every return, to see how far its number could move on another vintage of the data. **vintage**: the version of French's files on the download date, named by the CRSP cut French built them from. **provenance**: the record of the download date, the files' hashes and the vintage that every number was computed on. **basis point**: a hundredth of a percentage point. **standard error**: the standard deviation of an estimate; for a mean active return over 47 years, the realised tracking error divided by the square root of 47. **covariance matrix**: the table of the variances of the assets' returns and the covariances between them. **shrinkage**: pulling an estimate towards a simpler target; the shrinkage intensity is the weight on the target. **Bayes-Stein**: the rule of Jorion (1986) that shrinks each estimated mean towards one common value. **short-sale constraint**: the condition that no weight is negative; the budget constraint is the condition that the weights sum to one; a corner solution puts all wealth in one asset; the Lagrangian is the function whose minimisation under constraints the paper writes out. **factor**: a return series that many assets move with; here the market (Mkt-RF), size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum (Mom) factors of Fama and French and Carhart. **beta**: the slope of an asset's return on a factor's return, or on the benchmark's. **market capitalisation**: the number of a firm's shares times their price. **cap-weighted**: weighted by market capitalisation; said of the benchmark's industry weights. **value-weighted**: said of French's industry returns: each firm counts in proportion to its market capitalisation at the start of the month. **benchmark**: the index the portfolio tracks, the cap-weighted combination of the 49 industries. **tracking error**: the standard deviation of the monthly difference between the portfolio's return and the benchmark's, times the square root of 12; ex ante when forecast, realised when measured after the fact. **active return**: the portfolio's return minus the benchmark's; the active weight of an industry is its portfolio weight minus its benchmark weight. **tilt**: a constraint holding named industries below their benchmark weight, here Coal, Oil and Utilities at no more than half of it. **mandate**: the whole set of rules the extension's portfolio obeys: the tilt, the exclusion of tobacco and weapons, and a beta of one to the benchmark, within the active weight bound and the turnover cap. **covariance estimator**: a method that turns a window of returns into a covariance matrix; the minimum-variance portfolio built from it is the long-only or unconstrained portfolio with the smallest forecast variance. **principal component**: a weighted combination of the assets' returns that captures the largest variance not captured by the components before it. **bias statistic**: the standard deviation of the monthly active return divided by the same month's forecast (the forecast tracking error over the square root of twelve); 1 for a calibrated forecast, above 1 an under-forecast. **attribution**: the split of a month's active return into the contributions of the six factors, each the active exposure to the factor times its return, and a residual. **Jobson-Korkie**: the test of equal Sharpe ratios the paper uses, with the Memmel correction. **Student-t**: a distribution with fatter tails than the normal. **degrees of freedom**: the parameter of the Student-t distribution; the fewer, the fatter the tails. **oracle**: a rule given the true covariance of each month in advance, as a ceiling on what knowledge of changing risk is worth.

1. Introduction

Whether an optimised portfolio beats the rule that gives every asset the same weight is an old question in portfolio construction, and DeMiguel, Garlappi and Uppal (2009) gave it the answer that has become the reference point: on seven datasets and fourteen rules, none of the optimising rules beat 1/N (the rule that holds every asset at the same weight and rebalances to it each month) consistently out of sample, because the error in means and covariances

estimated from 120 months of returns costs more than the optimisation gains (Michaud, 1989; Jagannathan and Ma, 2003). An index-tracking fund with a sustainability mandate faces a narrower version of the question. Its mandate fixes most of the portfolio, since it must hold every industry within a bound of its benchmark weight, trade no more than a cap per month, hold named industries below their benchmark weight and exclude others, and the one estimate the optimiser needs is a covariance matrix (the table of variances and covariances of the assets' returns) that forecasts tracking error, the standard deviation of the difference between the portfolio's return and the benchmark's. The question such a fund poses is whether the choice of covariance estimator still matters once these constraints bind.

Dom, Howard, Jansen and Lohre (2024) ask this question of the global minimum-variance portfolio and find that shrinkage and covariance dynamics beat the sample estimator when the portfolio is unconstrained, that the advantage largely disappears once the portfolio is long-only with asset-level constraints and a transaction cost penalty, and that time-series dynamics remain relevant for volatility reduction. Their object is the portfolio with the smallest variance. An index fund's object is the smallest tracking error against a benchmark under a mandate, and whether their result carries to it is open on two counts. The constraints of a mandate are of a different kind: their long-only portfolio caps each stock at 1 per cent of the portfolio and penalises trading, while a mandate holds every industry within 2 percentage points of its benchmark weight and caps turnover at 2 per cent a month, a cap the benchmark's own movement can exhaust; and the quantity minimised is a difference of returns whose variance is more than two orders of magnitude below the market's, so the same estimation error in the covariance matrix can matter more or less in relative terms.

This note does two things. First, it replicates DeMiguel, Garlappi and Uppal (2009) on four of the five datasets the paper takes from French's library, the four with tables in the paper, against bands fixed before the code ran, and extends their test to the 261 months since their sample. Second, it asks their question of a tracking-error mandate on French's 49 industry portfolios. Each month from 1979-07 to 2026-08 an optimiser finds the long-only portfolio closest to a cap-weighted benchmark in forecast tracking error under an active weight bound of 2 percentage points, a one-way turnover cap of 2 per cent a month, a 50 per cent reduction of Coal, Oil and Utilities relative to the benchmark, the exclusion of tobacco and weapons, and a beta of one to the benchmark, with the forecast from one of five covariance estimates; the realised tracking error of each path (the month-by-month sequence of portfolios of one specification, each built on the data before its month) is the outcome. The design has no expected return in it, so it recovers the cost of the mandate in tracking error and in realised active return and the differences between estimators, and nothing about what a return forecast would add. A factor attribution then splits each path's active return into the contributions of six factors and a residual.

French's 49 industries are an informative setting for three reasons. The data are public and revised, so the replication's distance from the published figures can be read against a fragility test (the same rule rerun with small Gaussian noise added to every return) on the same data, and the vintage every number was computed on, the CRSP cut French built the files from, is recorded. The industries' weights in the market can be rebuilt from French's own firm counts and average firm sizes, and the rebuilt benchmark tracks his market return within 47 basis points a year (a basis point is a hundredth of a percentage point), so the fund tracks the market itself and the mandate's cost is measured against a known reference. And the industries' betas to the common factors (the slopes of their returns on the factors' returns) differ by about 0.5, so a tilt against three industries creates factor exposures large enough to measure and small enough, about 0.01 to 0.02, to be a realistic by-product of a sustainability mandate; an enhanced index fund with a 30 per cent reduction in carbon footprint relative to its benchmark runs at an ex-ante tracking error limit near 1 per cent (Robeco, 2024), the level this mandate reaches.

Empirically, the replication reproduces 29 of the 30 out-of-sample Sharpe ratios that carry a band of 0.03 and all 30 turnover figures within 25 per cent; the one miss, and the two datasets on which the top three rules exchange places, are reported at the resolution the band allows. In the 261 months since the paper, none of the seven optimising rules earns a higher Sharpe ratio than 1/N at the 5 per cent level on any of the four datasets. The one cell of the 32 that does is the value-weighted market on MKT/SMB/HML, which is not an optimising rule. Under the mandate the four daily estimators realise 84.7 to 93.4 basis points a year of tracking error and the monthly sample covariance 95.8, so the estimator still matters, by about a tenth of the tracking error, with the daily sample covariance and shrinkage the best of the five; every estimator under-forecasts its own tracking error, by 22 to 61 per cent. The tilt alone is associated with an active return of +2 to +7 basis points a year against a standard error of 12 to 13, and the attribution indicates that this zero is the sum of factor exposures that cost 9 to 13 basis points a year and industry returns that paid 12 to 20. What the

estimates exclude is a cost of the tilt in return beyond about 25 basis points a year, two standard errors, and a difference between the daily estimators beyond about a tenth of the tracking error.

2. Data and design

All data come from Ken French's library, downloaded at run time: six monthly files (the 10 and the 49 industry portfolios, the 25 portfolios formed on size and book-to-market, the three and the five Fama-French factors, and the momentum factor) and four daily files (the 49 industries and the three factor files). Nothing from the library is stored in the repository; a provenance record keeps the download date, the SHA-256 of each zip and the CRSP vintage, 202608 for every number here, and the tables the notebooks write, which every figure in this note rests on, are committed in the repository's `results/202608/` folder with that record. The paper takes five of its seven datasets from the library and prints tables for four, because the fifth, FF-3-factor, gives results almost identical to FF-1-factor (its Table 2); the four are Industry (the 10 industries and the market, 11 assets), MKT/SMB/HML (the three factors), FF-1-factor (20 size-and-book-to-market portfolios and the market, the five largest-size portfolios dropped as in the paper's footnote 24) and FF-4-factor (the same 20 with the market, size, value and momentum factors). Returns are monthly excess returns over the one-month Treasury bill.

The replication runs the paper's nine rules on each dataset through its rolling evaluation: an estimation window of 120 months, the window's sample moments fed to the rule, the weights held for the following month, the window moved forward one month, over the paper's sample of 1963-07 to 2004-11, which leaves 377 out-of-sample months. The rules are 1/N; sample-based mean-variance, which plugs the window's mean and covariance into Markowitz's formula; Bayes-Stein, which shrinks each estimated mean towards a common value (Jorion, 1986); minimum variance, which uses the covariance alone; the value-weighted market; and the three short-sale-constrained versions (weights that may not be negative) of mean-variance, Bayes-Stein and minimum variance, with the generalised minimum-variance rule that holds every weight at or above 1/(2N) (Jagannathan and Ma, 2003). The measures are the out-of-sample Sharpe ratio (the mean monthly excess return divided by its standard deviation), the certainty-equivalent return (the mean minus half the variance, at the paper's risk aversion of one) and turnover (the sum of absolute weight changes at each rebalance, averaged). Three criteria were fixed before the code ran and written into a constants file with their sources: each out-of-sample Sharpe ratio within 0.03 of the published value, the ranking of the top three rules within each dataset preserved, and each turnover figure within 25 per cent of the published one. One specification the paper leaves open was settled by its own footnote and recorded: its Lagrangian for the constrained rules shows the short-sale constraint alone, and its footnote 22 says those rules produce corner solutions with all wealth in one asset, which happens only when the budget constraint sits inside the optimisation; both readings were run, the one that reproduces the tables is used and the other is kept as a named function.

One decision was changed after a run, on evidence recorded with it. The unconstrained mean-variance rule's published turnover is 606,594 times that of 1/N on the Industry dataset and 10,466 times on FF-1-factor, figures dominated by months in which the sample covariance is near-singular (a matrix whose inverse multiplies small errors by thousands), and under ten basis points of noise added to every return the rule's Sharpe ratio moved by 0.12 to 0.25 on three datasets with the published figure inside the range, while the 1/N rows sit within 0.001 of the paper on three datasets. The Sharpe band therefore applies only where the paper's own relative turnover is below 100, the level the turnover band already used, and the other cells are reported beside the published figure with no verdict. This leaves 30 gated Sharpe cells and 30 gated turnover cells of the 36 of each.

Four parts of the study go beyond the paper's tables. The paper's Proposition 1, the estimation window at which sample-based mean-variance starts to beat 1/N on average, is checked against the 13 values the paper states, and its simulation (Table 6) is rerun on one-factor markets of 10, 25 and 50 assets. Two further simulated markets test what the comparison rests on: Student-t shocks with 5 and 10 degrees of freedom, and GARCH(1,1) volatility, under which months are no longer independent and the correlations between assets move. The nine rules are then run on every month the files carry, to 2026-08, and the 261 months since the paper are compared with the paper's period; the difference between a rule's Sharpe ratio and 1/N's is tested with the Jobson-Korkie statistic as the paper corrects it, and a p-value is the probability of a difference at least this large if the two rules were equally good.

The extension's benchmark is the cap-weighted combination of the 49 industries, each industry's weight its number of firms times its average firm size from French's own file, divided by the total. Over the 686 months from 1969-07 the benchmark's excess return tracks French's market excess return (Mkt-RF) within 46.7 basis points a year of tracking error against a gate of 50 fixed before the run, with a correlation of 0.9996; the gap comes from French's end-of-June formation, under which a firm listed since June is in the market and in no industry until the next July, and a direct test on the CRSP monthly file (run where that file is, with only aggregates kept) accounts for 84 per cent of the gap's variance.

Five covariance estimates are compared, each estimated on the data before the month. Four are built from three years of daily excess returns, about 756 days: the sample covariance; Ledoit-Wolf shrinkage, which pulls the sample covariance towards a scaled identity matrix with the intensity the data choose (Ledoit and Wolf, 2004); a six-factor model, in which each industry's return is regressed on the market, size, value, profitability, investment and momentum factors and the covariance is the factors' covariance carried through the betas plus a diagonal of residual variances; and a principal-component model, which takes the benchmark's return as its first component and the first four principal components of what the benchmark leaves (the directions of largest remaining variance) as the others, plus a diagonal. The fifth is the sample covariance of the last 120 monthly returns, the paper's convention. The principal-component model takes the market as its first component by a consequence fixed in advance: the first component of the 49 industries' monthly excess returns over 1969-07 to 2026-08 explains 55 per cent of their variance and has a correlation of 0.956 with the benchmark's excess return, against a pre-committed expectation of 0.95. Two further portfolios run beside the five: a robust portfolio whose weights minimise the largest forecast tracking error across the five covariances, and a RiskMetrics portfolio from an exponentially weighted covariance with the one-year half-life of Dom, Howard, Jansen and Lohre (2024), on five years of daily returns.

Each month the optimiser solves a convex quadratic programme: the weights that minimise the forecast tracking error against the benchmark under the constraint set in force, starting from the previous month's weights after the month's returns have moved them. Four cumulative constraint sets are run. C0 is long-only; C1 adds the active weight bound, each industry within 2 percentage points of its benchmark weight; C2 adds the one-way turnover cap of 2 per cent a month, half the sum of absolute weight changes, relaxed to the smallest feasible turnover in the months in which the benchmark itself moves more than the cap allows; C3 adds the mandate, Coal, Oil and Utilities at no more than half their benchmark weight, tobacco and weapons at zero, and a beta of one to the benchmark computed from the covariance in use. The tilt alone, C2 plus the 50 per cent reduction, runs as an eighth set. Under C0 and C1 the benchmark itself is feasible and the portfolio reproduces it. Each path runs from 1979-07, when both the 120-month and the three-year windows are available, to 2026-08, 566 months, and 28 specifications (seven covariance estimates times four sets) plus the seven tilt-alone paths are evaluated on the realised tracking error, the mean forecast, the bias statistic (the standard deviation of the monthly active return divided by the same month's forecast, the forecast tracking error over the square root of twelve; 1 for a calibrated forecast), the active return (the portfolio's return minus the benchmark's, times 12), one-way turnover and trading costs at 50 and 100 basis points per unit of turnover. The standard error of a path's active return is its realised tracking error over the square root of 47 years, and the cost of the mandate against the tilt alone is a difference of means on the same months with its own standard error.

The attribution rebuilds the fourteen paths under C3 and under the tilt alone with their weights and splits each month's active return into six contributions and a residual. Each industry's betas to the six factors come from one regression of its daily excess returns on the factors over the three years before the month, so the betas are known when the month begins; the active exposure to a factor is the sum over industries of the active weight (the portfolio weight minus the benchmark weight) times the industry's beta; a contribution is the exposure times the factor's return in the month; the residual is the active return minus the six contributions, the part the industries' own returns account for. Two uses of beta sit side by side by design: the optimiser's beta-neutrality rule uses each industry's beta to the benchmark from the covariance it works with, and the attribution uses the betas to the six factors, because the report splits the return across the factors a client knows; the benchmark and the market factor move almost identically, so the two market betas differ by about 0.005 at most.

Every expectation of the extension was written before its code ran, in each notebook's introduction and in the constants file, and the specification count was fixed in advance. Three design changes and two additions were made after a first run; each is labelled in the notebooks with its date and had its expectation written before its code. Beta neutrality joined

the mandate when the hedge built from the monthly covariance was found to leave a beta of 1.007 to 1.010 standing; the exclusions joined it when the hedge bought tobacco; the robust portfolio was added when the hedge differed across estimators by 1 to 5 basis points; and the RiskMetrics portfolio and the calibration of the forecasts on each path's own history were added to the evaluation. The cost of the three changes against the design they replaced is reported in Section 3.

3. Results

3.1 The replication

Table 1 reports the out-of-sample Sharpe ratios beside the paper's, and the paper's Table 3 reproduces. Of the 30 gated cells, 29 are within the band of 0.03 and 27 within 0.01; the 1/N cells, which carry no estimation, are within 0.001 of the paper on three datasets and 0.011 on MKT/SMB/HML, whose three assets are the factor series themselves and have been revised more than the portfolios, and the certainty-equivalent returns of the four constrained rules are within 0.0004 of the paper's on all 16 cells and within 0.0002 on 15. The Bayes-Stein shrinkage factor averages 0.33 on FF-4-factor and 0.68 on MKT/SMB/HML against the paper's 0.32 and 0.66, so the shrinkage is implemented as Jorion specified it. The paper's reading of its table (its fourth and fifth observations) also reproduces, cell for cell in sign: constrained mean-variance and constrained Bayes-Stein have a lower Sharpe ratio than 1/N on Industry and MKT/SMB/HML and a higher one on the two size-and-value datasets, and constrained minimum variance has a higher one than 1/N on Industry, MKT/SMB/HML and FF-4-factor and a lower one on FF-1-factor. Of the three differences the paper reports as significant at the 5 per cent level, two are significant here (constrained Bayes-Stein above 1/N on FF-1-factor, p = 0.04; constrained minimum variance above 1/N on FF-4-factor, p = 0.0004) and the third, constrained mean-variance on FF-1-factor, has p = 0.06.

One gated cell misses. Minimum variance on FF-4-factor is 0.033 against the paper's -0.018, and the fragility test does not account for it: ten basis points of noise move the cell across a range of 0.08 that does not reach the published value. The in-sample mean-variance row, which carries no verdict, points to the same cause, since it sits 0.04 to 0.06 below the published figure on the two size-and-value datasets and within 0.01 on Industry, which indicates that the 202608 vintage of the 25 size-and-book-to-market portfolios differs from the vintage the paper used in a way the industry file does not. The ranking criterion passes on Industry and FF-4-factor and fails strictly on the other two, where the rules that change places have published Sharpe ratios 0.002 to 0.007 apart; a band of 0.03 cannot distinguish an exchange of places at that resolution from a replication error.

Table 2 reports turnover, and all 30 gated cells are within 25 per cent of the published figures, most within 10 per cent. The unconstrained mean-variance and Bayes-Stein rules on Industry, FF-1-factor and FF-4-factor are 2,347 and 606,594 times 1/N in the two columns of the first cell and carry no verdict; the five worst months of the rule on Industry hold positions summing to 200 to 4,500 times the portfolio's value, and a difference of orders of magnitude between two vintages in such a cell measures the fragility of the rule; the constrained rules, which trade 1 to 19 times 1/N, are within 25 per cent of the paper's figure in every cell.

The paper's analytical result and its simulation reproduce. Proposition 1's critical window, the estimation window at which sample-based mean-variance starts to beat 1/N on average, matches all 13 values the paper states, and 77 of the 81 cells of its Table 6 are within 0.03, with the sign pattern of mean-variance against 1/N reproduced on every market. In the fat-tailed markets the estimated rules lose 0.002 in Sharpe ratio at 5 degrees of freedom and nothing detectable at 10, which is consistent with the cost of estimation sitting in the means, since the sampling error of a mean does not depend on the tails. In the market with clustered volatility the comparison does not change, and an oracle that knows each month's true covariance gains 0.003 to 0.005 in Sharpe ratio over a rule that holds the long-run optimal weights, about 0.2 per cent of return a year on a 15 per cent volatility, which bounds what any estimate of changing risk can add to a rule that only decides how to split wealth between assets.

Table 1. Out-of-sample Sharpe ratios, 1973-07 to 2004-11 (377 months), this study beside DeMiguel, Garlappi and Uppal (2009, Table 3)

	Industry		*MKT/SMB/HML*		*FF-1-factor*		*FF-4-factor*	
Rule	*this study*	*DGU*	*this study*	*DGU*	*this study*	*DGU*	*this study*	*DGU*
mean-variance, in sample	0.2034[b]	0.2124	0.2608[b]	0.2851	0.4509[b]	0.5098	0.4999[b]	0.5364
1/N	0.1353	0.1353	0.2347	0.2240	0.1613	0.1623	0.1745	0.1753
mean-variance	-0.0494[b]	0.0679	0.2135	0.2186	0.0513[b]	0.0128	0.1650[b]	0.1841
Bayes-Stein	-0.0383[b]	0.0719	0.2541	0.2536	0.0620[b]	0.0138	0.1672[b]	0.1791
minimum variance	0.1537	0.1554	0.2574	0.2493	0.2496	0.2778	0.0326[a]	-0.0183
value-weighted market	0.1163	0.1138	0.1163	0.1138	0.1163	0.1138	0.1163	0.1138
mean-variance, constrained	0.0627	0.0678	0.1093	0.1084	0.1963	0.1977	0.1978	0.2024
Bayes-Stein, constrained	0.0831	0.0819	0.1480	0.1514	0.1974	0.1955	0.2064	0.2062
minimum variance, constrained	0.1445	0.1425	0.2574	0.2493	0.1554	0.1546	0.3649	0.3580
minimum variance, generalised constraint	0.1454	0.1451	0.2552	0.2467	0.1625	0.1615	0.3080	0.3028

Each cell is the Sharpe ratio of the rule's monthly out-of-sample excess returns with a 120-month estimation window, as in the paper. Thirty cells carry a band of 0.03 fixed before the code ran; 29 are within it. [a] The one miss: 0.0326 against -0.0183. [b] No verdict, by a decision recorded with its evidence in the constants file: the in-sample row, and the unconstrained mean-variance and Bayes-Stein rules on the datasets where the paper's own turnover figure for the rule exceeds 100 times that of 1/N, because ten basis points of noise added to every return move those cells by 0.12 to 0.25 with the published figure inside the range.

Table 2. Turnover, 1973-07 to 2004-11: 1/N in absolute terms (the fraction of the portfolio traded per month, counting purchases and sales), every other rule as a multiple of 1/N, this study beside DeMiguel, Garlappi and Uppal (2009, Table 5)

	Industry		*MKT/SMB/HML*		*FF-1-factor*		*FF-4-factor*	
Rule	*this study*	*DGU*	*this study*	*DGU*	*this study*	*DGU*	*this study*	*DGU*
1/N	0.0221	0.0216	0.0237	0.0237	0.0164	0.0162	0.0200	0.0198
mean-variance	2,347[b]	606,594	2.50	2.83	10,112[b]	10,466	678.03[b]	3,553
Bayes-Stein	1,952[b]	10,621	1.52	1.85	5,516[b]	11,796	795.93[b]	3,418
minimum variance	20.99	21.65	1.02	1.11	44.95	45.47	6.16	6.83
value-weighted market	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
mean-variance, constrained	7.70	7.17	3.86	4.12	19.07	17.53	16.46	13.82
Bayes-Stein, constrained	6.63	7.22	3.45	3.65	16.11	17.32	13.65	13.07
minimum variance, constrained	2.34	2.58	1.02	1.11	4.20	3.93	1.63	1.76
minimum variance, generalised constraint	1.49	1.52	1.00	1.09	1.95	1.78	1.66	1.70

Thirty cells carry a band of 25 per cent, relative, fixed before the code ran; all 30 are within it. [b] No verdict: the paper's figure exceeds 100 times that of 1/N, and the cell moves by orders of magnitude when a single month's covariance is near-singular. The value-weighted market trades nothing by construction.

3.2 The months since the paper

Table 3 reports the Sharpe ratios on the paper's period and on the 261 months from 2004-12 to 2026-08. In the new period one of the 32 rule-dataset cells beats 1/N at the 5 per cent level, the value-weighted market on MKT/SMB/HML, and five cells fall below 1/N at that level, all five of them minimum-variance rules on the two factor datasets. Minimum variance, the best optimising rule of the paper's period, falls to -0.015 on MKT/SMB/HML and -0.034 on FF-4-factor against 0.113 and 0.148 for 1/N; its weights load on the size and value factors, whose premia were small or negative over these years, so the fall is consistent with the premia it loaded on and with no change in the rule. On the whole period from 1973-07 the constrained rules beat their unconstrained versions in 11 of 12 comparisons and 1/N beats sample-based mean-variance on every dataset. These estimates do not distinguish most rules from 1/N: a monthly Sharpe ratio measured on 261 months has a standard error of about 0.06, so a difference between two rules has to reach about 0.12 before it is distinguishable from chance, and most differences in the new period are smaller.

Table 3. Out-of-sample Sharpe ratios on the paper's period and on the months since, on the 202608 vintage of French's files

	1973-07 to 2004-11 (377 months)				*2004-12 to 2026-08 (261 months)*			
Rule	*Industry*	*MKT/SMB/HML*	*FF-1-factor*	*FF-4-factor*	*Industry*	*MKT/SMB/HML*	*FF-1-factor*	*FF-4-factor*
1/N	0.135	0.235	0.161	0.174	0.191	0.113	0.148	0.148
mean-variance	-0.049	0.213	0.051	0.165	0.042	0.176	0.176	0.102
Bayes-Stein	-0.038	0.254	0.062	0.167	0.053	0.158	0.208	0.101
minimum variance	0.154	0.257	0.250	0.033	0.168	-0.015	0.218	-0.034
value-weighted market	0.116	0.116	0.116	0.116	0.191	0.191	0.191	0.191
mean-variance, constrained	0.063	0.109	0.196	0.198	0.174	0.189	0.151	0.143
Bayes-Stein, constrained	0.083	0.148	0.197	0.206	0.190	0.188	0.159	0.146
minimum variance, constrained	0.144	0.257	0.155	0.365	0.210	-0.009	0.181	0.038
minimum variance, generalised constraint	0.145	0.255	0.163	0.308	0.210	0.042	0.165	0.149

The paper's period on this vintage reproduces Table 1 (29 of 30 gated Sharpe cells, 30 of 30 turnover cells). In the new period one of the 32 rule-dataset cells differs from 1/N in its favour at the 5 per cent level (the value-weighted market on MKT/SMB/HML, p = 0.04); five differ against 1/N at that level, all of them minimum-variance rules on the two factor datasets. A monthly Sharpe ratio measured on 261 months has a standard error of about 0.06, so a difference between two rules has to reach about 0.12 before it is distinguishable from chance.

3.3 The estimators under the mandate

Without constraints the estimator matters as Dom, Howard, Jansen and Lohre (2024) find. The minimum-variance portfolio built from the sample covariance of 120 monthly returns realises 14.1 per cent volatility a year over the 566 months against 12.1 to 12.8 per cent for the three structured estimators; on three years of daily returns the sample covariance is as good as any (11.9 per cent against 11.9 to 13.5); and the long-only constraint brings every estimator at either frequency to within 0.2 points of the others, 12.1 to 12.3 per cent. At 50 basis points per unit of turnover the unconstrained portfolios pay 1.3 to 4.6 per cent a year in trading and earn less net of cost than the long-only ones in all eight variants.

Table 4 reports the mandate. The constraints without the mandate cost little in tracking error: under C0 and C1 every path reproduces the benchmark, and under C2 the turnover cap alone realises 6.3 to 6.5 basis points a year, from the one month in which the benchmark moved more than the cap allowed and the months the portfolio took to catch up. The mandate realises 84.7 basis points a year with the daily sample covariance, 84.8 with shrinkage, 91.5 with the principal-component model, 93.4 with the factor model, 95.8 with the monthly sample covariance, 87.4 for the robust portfolio and 84.3 for RiskMetrics; the tilt alone realises 79.7 to 90.9 on the same seven, so the tilt is 94 to 95 per cent of the mandate's tracking error and beta neutrality with the exclusions adds 4 to 5 basis points. The range across the four daily estimators is 8.8 basis points on a level of about 90, so the differences the constraints leave are about a tenth of the tracking error. The ranking is the one the unconstrained comparison on daily data gave: on three years of daily returns the sample covariance and shrinkage are almost the same estimate (a shrinkage intensity of 0.012) and the best, the two structured models pay 7 to 9 basis points a year for assuming that industries share nothing beyond their factors, and the monthly covariance pays for its noise. The gain from time dynamics that Dom, Howard, Jansen and Lohre (2024) report for the global minimum-variance portfolio does not appear under the mandate: RiskMetrics realises 84.3 against the sample covariance's 84.7, inside the resolution of this comparison.

Every estimator under-forecasts its own tracking error. The bias statistic is 1.22 to 1.39 for the four daily estimators, 1.61 for the monthly covariance and 1.09 for the robust portfolio, whose worst-case forecast is the best calibrated by construction. The direction was foreseeable, since an optimiser that minimises a forecast picks the weights for which the forecast is lowest relative to the truth (Michaud, 1989) and the same estimators' forecasts for the individual industries, where no optimiser chooses, carry a bias statistic of 1.10 on daily data; the size was not, and by decade the bias is largest in 1979 to 1989, when the oil shock fell on a portfolio tilted away from Oil, which held 15 per cent of the benchmark in 1979. Calibrating each month's forecast by the bias of the same path over the previous 60 months, with nothing from the future, moves the daily estimators to 1.11 to 1.17 and the monthly covariance from 1.61 to 1.12, so the persistent part of the bias from 1990 on is repaired by the path's own history and the remaining 11 to 17 per cent is the part a 60-month window cannot see.

The mandate's return is in the last columns of Table 4. The tilt alone is associated with an active return of +1.8 to +6.9 basis points a year across the seven estimates, against a standard error of 11.6 to 13.2, so the tilt's cost in return is indistinguishable from zero and the estimates exclude a cost beyond about 25 basis points a year. The full mandate is associated with -1.0 to -6.9 against a standard error of 12.3 to 13.9. The difference between the two, the cost of beta neutrality and the two exclusions on the same months, is -7.9 to -10.4 basis points a year with a standard error of 3.4 to 4.9 and the same sign under all seven estimates, so this part of the mandate has a cost the data can see. The mandate's portfolios turn over 0.86 to 1.54 per cent a month one way against the benchmark's 0.39, hold 41 to 45 of the 49 industries and an effective number of about 19 against the benchmark's 18.9, and at 50 basis points per unit of turnover pay 5.1 to 9.2 basis points a year in trading against the index fund's 2.3. The three design changes cost the daily estimators 2 to 7 basis points of forecast tracking error at the three months checked and 4 to 5 basis points realised over the full period, and the robust portfolio sits 0.1 to 1.1 basis points above each estimator's own best forecast at those months.

Table 4. The seven covariance estimates under the mandate, 1979-07 to 2026-08 (566 months): realised tracking error, forecast calibration, active return and trading

Covariance estimate	*Realised tracking error, bp a year*	*Mean forecast, bp a year*	*Bias statistic*	*Bias, calibrated*	*Active return, mandate, bp a year (s.e.)*	*Active return, tilt alone, bp a year (s.e.)*	*Mandate minus tilt alone, bp a year (s.e.)*	*One-way turnover, % a month*	*Trading cost at 50 bp, bp a year*
Sample covariance, 3 years daily	84.7	59.8	1.26	1.11	-6.9 (12.3)	+1.8 (11.7)	-8.7 (3.7)	1.31	7.8
Ledoit-Wolf shrinkage, 3 years daily	84.8	61.9	1.22	1.11	-6.4 (12.4)	+2.8 (11.7)	-9.1 (3.7)	1.23	7.4
Six-factor model, 3 years daily	93.4	60.8	1.37	1.17	-1.0 (13.6)	+6.9 (12.9)	-8.0 (3.4)	0.86	5.1
Principal components, 3 years daily	91.5	58.3	1.39	1.15	-5.8 (13.3)	+3.9 (12.6)	-9.7 (3.8)	0.96	5.8
Sample covariance, 120 months	95.8	55.0	1.61	1.12	-2.4 (13.9)	+6.6 (13.2)	-9.0 (4.9)	1.54	9.2
Robust (worst case of the five)	87.4	67.8	1.09	1.06	-4.8 (12.7)	+5.6 (12.0)	-10.4 (4.3)	1.06	6.4
RiskMetrics (exponentially weighted), 5 years daily	84.3	60.4	1.23	1.08	-5.3 (12.3)	+2.7 (11.6)	-7.9 (3.8)	1.44	8.6

Tracking error is the standard deviation of the monthly active return times the square root of 12; the bias statistic is the standard deviation of the monthly active return divided by the same month's forecast (the forecast tracking error over the square root of twelve), 1 for a calibrated forecast, so it weights each month by its own forecast and differs from the ratio of the realised column to the forecast column; the calibrated column rescales each month's forecast by the bias of the same path over the previous 60 months, with nothing from the future. The standard error of an active return is its realised tracking error over the square root of 47 years; the standard error of the difference is computed on the paired monthly differences. One-way turnover is half the sum of absolute weight changes at a rebalance; the benchmark itself turns over 0.39 per cent a month, which at 50 basis points per unit of turnover costs an index fund 2.3 basis points a year.

3.4 Where the active return came from

Table 5 reports the attribution. The identity holds to 2e-18 on every path, and the rebuilt paths reproduce the realised tracking errors of Table 4 within 0.1 basis point. Beta neutrality shows: the mean active exposure to the market factor is between -0.005 and +0.001 on all fourteen paths, under the tilt alone, where it is the optimiser's own work, as under the mandate, where it is a constraint. The exposure matters because it is the most expensive position a tracking-error mandate can hold: an exposure of 0.05 times the market's volatility of 15.6 per cent a year is 78 basis points of tracking error, most of the 100 an enhanced index fund allows, and this portfolio makes no forecast of the market's direction that would justify spending them.

The tilt leaves five exposures behind, with the same signs under every estimate. Under the mandate the portfolio holds +0.010 to +0.018 more of the size factor than the benchmark and -0.010 to -0.015 of value, -0.008 to -0.012 of profitability, -0.005 to -0.008 of investment and -0.004 to -0.005 of momentum. The signs follow from what the tilt removes. Averaged over the evaluation months Oil's betas are -0.16 to size, +0.27 to value, +0.12 to profitability, +0.45 to investment and -0.07 to momentum, and Utilities' -0.10, +0.35, +0.02, +0.17 and +0.02, against the benchmark's +0.05, 0.00, -0.01, +0.01 and -0.04; holding half of two industries that behave like large, cheap, profitable, low-investment firms and giving the weight to industries with the benchmark's betas would move the exposures by +0.010 to size, -0.020 to value, -0.010 to profitability, -0.012 to investment and -0.003 to momentum, and the paths under the tilt alone show +0.008 to +0.017, -0.010 to -0.016, -0.004 to -0.009, -0.003 to -0.006 and -0.003 to -0.005, the differences

being the substitutes' own betas. The exposures are small because the active weight bound is: 2 percentage points per industry times betas that differ by about 0.5 across industries gives exposures of about 0.01, and they were largest in the first decade, when Oil held 15 per cent of the benchmark and the tilt removed 7.6 points of it.

The six factors explain 14 to 28 per cent of the active return's variance across the fourteen paths, so most of the tracking error is the industries' own returns, the risk the factor models of Section 2 assume away between industries. The attribution also splits the tilt's zero return into two parts of opposite sign: under the tilt alone the factor contributions sum to -9.3 to -13.1 basis points a year, about two standard errors from zero, and the residual to +12.2 to +19.7, one to two standard errors. The exposures the tilt created cost what their signs and the factors' mean returns over these years imply, since value, profitability, investment and momentum all earned positive premia on average and the portfolio was short of each, and the industries the optimiser bought in place of Oil and Utilities earned more than Oil and Utilities did over the 47 years; the first part follows from the exposures and the factors' returns and appears under every estimator, the second is the history of three industries. Of the mandate's cost against the tilt alone, 20 to 40 per cent comes through the factors and 60 to 80 per cent through the industries' own returns, and the exclusions' factor part, the five factors other than the market, is -1.3 to -2.9 basis points a year against the -2.9 that their mean weight of 1.3 per cent and their betas imply, within the 2 basis points fixed in advance. By decade, for the sample covariance's path under the mandate, the factor part was -61 basis points a year in 1979 to 1989 and within 13 of zero in every decade since, so the exposures' cost is concentrated in the years when the tilt was largest.

Table 5. Factor attribution of the active return, 1979-07 to 2026-08: mean active exposures, contributions and residual

Covariance estimate	*Mkt-RF*	*SMB*	*HML*	*RMW*	*CMA*	*Mom*	*Factor contributions, bp a year (s.e.)*	*Residual, bp a year (s.e.)*	*Factor share*
Under the mandate									
Sample covariance, 3 years daily	-0.001	+0.013	-0.014	-0.011	-0.005	-0.004	-13.8 (5.5)	+6.9 (10.9)	0.22
Ledoit-Wolf shrinkage, 3 years daily	-0.001	+0.014	-0.015	-0.011	-0.005	-0.004	-13.8 (5.6)	+7.5 (10.9)	0.23
Six-factor model, 3 years daily	+0.001	+0.012	-0.010	-0.008	-0.005	-0.004	-10.9 (5.4)	+9.9 (12.6)	0.14
Principal components, 3 years daily	0.000	+0.010	-0.015	-0.012	-0.008	-0.005	-15.1 (6.2)	+9.3 (11.9)	0.20
Sample covariance, 120 months	-0.005	+0.018	-0.015	-0.008	-0.006	-0.005	-16.7 (6.6)	+14.3 (12.1)	0.25
Robust (worst case of the five)	-0.001	+0.013	-0.014	-0.009	-0.005	-0.004	-14.1 (5.7)	+9.3 (11.2)	0.22
RiskMetrics (exponentially weighted), 5 years daily	-0.001	+0.013	-0.015	-0.011	-0.005	-0.004	-13.9 (5.6)	+8.6 (10.9)	0.21
Under the tilt alone									
Sample covariance, 3 years daily	0.000	+0.011	-0.014	-0.008	-0.003	-0.003	-10.5 (5.1)	+12.2 (10.3)	0.23
Ledoit-Wolf shrinkage, 3 years daily	0.000	+0.012	-0.015	-0.008	-0.004	-0.003	-10.4 (5.2)	+13.2 (10.2)	0.24
Six-factor model, 3 years daily	+0.001	+0.010	-0.010	-0.005	-0.003	-0.004	-9.3 (5.0)	+16.2 (12.0)	0.14
Principal components, 3 years daily	0.000	+0.008	-0.015	-0.009	-0.006	-0.005	-12.4 (6.0)	+16.3 (11.1)	0.22
Sample covariance, 120 months	-0.004	+0.017	-0.016	-0.004	-0.005	-0.004	-13.1 (6.3)	+19.7 (11.2)	0.28
Robust (worst case of the five)	-0.001	+0.012	-0.013	-0.007	-0.004	-0.004	-11.1 (5.6)	+16.8 (10.2)	0.27
RiskMetrics (exponentially weighted), 5 years daily	0.000	+0.012	-0.014	-0.008	-0.003	-0.003	-10.7 (5.1)	+13.4 (10.3)	0.22

The active exposure to a factor is the sum over industries of the active weight times the industry's beta to the factor, the betas from a regression of each industry's daily excess returns on the six factors over the three years before the month; its mean over the 566 months is shown. A contribution is the exposure times the factor's return in the month; the residual is the active return minus the six contributions, and the identity holds to 2e-18. The factor share is one minus the variance of the residual over the variance of the active return. Mkt-RF is the market factor, SMB size, HML value, RMW profitability, CMA investment and Mom momentum (Fama and French, 2015; Carhart, 1997).

4. Discussion

The natural comparison is with the two papers the design rests on. On DeMiguel, Garlappi and Uppal (2009) the replication indicates that their tables reproduce on a vintage of French's data built in 2026 against a sample that ends in

2004-11, with the misses where the size-and-book-to-market file was revised and where the rule itself is fragile, and that their conclusion holds on the 261 months since: no optimising rule beats 1/N by a margin these months can distinguish from chance, and the rule that came closest in their period, minimum variance, fell with the premia it loaded on. On Dom, Howard, Jansen and Lohre (2024) the extension indicates that their finding carries to a tracking-error mandate in part. The unconstrained comparison reproduces their pattern, structure and shrinkage beating the sample covariance on monthly data and the long-only constraint closing the gap to 0.2 points. Under the mandate the estimators differ by about a tenth of the tracking error, 8.8 basis points on 90, so the constraints shrink the differences without removing them, and the time dynamics that remained relevant in their minimum-variance setting do not show here, RiskMetrics realising 84.3 basis points against the sample covariance's 84.7.

For the holder of such a mandate the estimates therefore suggest three things. The choice among estimators built from three years of daily returns is worth about 9 basis points a year of tracking error, with the sample covariance and shrinkage at the low end and the two structured models at the high end, and the monthly sample covariance a further 2 to 11 behind; the forecast any of them produces understates the realised tracking error, by 22 to 39 per cent for the daily estimators and 61 for the monthly covariance, and a calibration on the path's own history removes most of the understatement. The tilt itself, a 50 per cent reduction of three industries, is associated with no cost in return the data can see and with a cost in tracking error of 80 to 91 basis points a year, close to the 1 per cent an enhanced index fund budgets. And the exposures the tilt leaves behind are unintended, since nobody chose them, and uncompensated in the sense that nothing in the mandate pays for them: at the factors' mean returns the value exposure of the tilt-only paths costs about 4 basis points a year against a yearly swing of about 15, variance added for little expected return; the attribution indicates that the six exposures together cost 9 to 13 basis points a year over these 47 years and that the industries bought in their place paid more, a combination that need not recur.

At the same time, the design has limits. First, the assets are industry portfolios, so the active weight bound, the exclusions and the attribution operate on 49 value-weighted groups of firms and the findings are about industries; a stock-level mandate would have hundreds of positions and an active weight bound a fraction of this one, and whether the differences between estimators are a tenth of the tracking error there is a question these data cannot answer. Second, the study holds no view on expected returns by design, because the replication located the cost of estimation in the means; the cost of the mandate is measured in realised return and tracking error, and what a return forecast would add to or take from it is outside the design. Third, one of the three post-run design changes, beta neutrality, was made on evidence from the same data it is evaluated on; its cost is labelled and reported, and the paths it replaced are kept as the tilt alone, so the reader can see both. Fourth, the attribution's betas are estimates from three years of daily returns and carry sampling error, so each exposure of about 0.01 is a sum of 49 estimated betas times active weights of at most 0.02, which is why the exposures are reported as means over 566 months and their month-by-month values are not read. Finally, French revises history, so a rerun on a later vintage will reproduce the comparisons and not every digit, and the fragility test is the measure of how far a digit can move.

What would settle the open questions is a stock-level panel with the same mandate, which would require a point-in-time constituent list and market capitalisations, and a dynamic estimator built for tracking error, which would require a model of the covariance's time variation estimated on the active return itself. Against this evidence, the claim is a narrow one: on French's 49 industries, under a mandate that holds the portfolio within 2 percentage points of its benchmark, trades no more than 2 per cent a month and reduces three industries by half, the covariance estimator still matters by about a tenth of the tracking error, the daily sample covariance and shrinkage are the best of the five, every estimator understates its own tracking error, and the tilt's cost in return is indistinguishable from zero while the factor exposures it creates are not.

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Corrected 10 November 2026. The summary in section 1 said that one of the 32 rule-dataset cells beats 1/N in the 261 months since the paper, without naming the cell, so it read as if an optimising rule had. It now says that none of the seven optimising rules earns a higher Sharpe ratio than 1/N at the 5 per cent level on any of the four datasets and that the one cell that does is the value-weighted market on MKT/SMB/HML, which is not an optimising rule, as section 3.2 and Table 3 report.